LAB 1.3 Logistic Population Models with Harvesting

Abstract

The laboratory experiment deals with the modified logistic equation for a fish population with constant harvesting and the periodic harvesting. Integrals and solution curves were done to analyze the nature of equilibrium points and solutions behavior. In constant harvesting the maximum limit of harvesting of fish will occur at 0.25; exceeding the value would cause extinction to the population for the rate of population could not catch up to the rate of harvesting. In the periodic harvesting, since the equation is non-autonomous, the effect of the initial condition when $a=a_1$ is a decrease in the population of the fish per season of harvesting while a rapid drop when $a=a_2$.

I. INTRODUCTION

Population growth is often modeled by first-order differential equations. An example of this is the Logistic equation.

The logistic model is based on the exponential growth and decay model

$$\frac{dp}{dt} = kp \tag{1}$$

where k is the per capita growth rate and P is the size of the population but it includes an overcrowding term, or non-constant growth rate, that reflects the limitations on growth due to the scarcity of resources and living space. The new term is proportional to the square of the population, so the equation becomes

$$\frac{dP}{dt} = kP - k\frac{P^2}{N} \tag{2}$$

K is the steady state, or carrying capacity, for the population in a particular habitat. The term $k\frac{P^2}{N}$ is a morality term in which P^2 represents competitive encounters between members of the population.

An application of the Logistic Equation is on population harvesting. Population harvesting is the removal of constant number of individual from a population during each time period. Constant harvesting is where a fixed number of animals were removed each year, while periodic harvesting is usually thought of as a sequence of periodic closure and openings of different fishing grounds (Blanchard, Devaney, & Hall, 2010).

II. OBJECTIVES

The experiment aims to understand the Logistic growth model with different types "harvesting" for a fish population. The specific objectives are as follows:

1. Determine the effect on the fish population for various initial condition if $a = a_1$.
2. Determine the effect on the fish population for various initial condition if $a = a_2$.
3. Describe the parameter used in the model using various initial conditions.

III. METHODOLOGY

In this lab, logistic models of population growth that have been modified to include terms that account for "harvesting" were considered. In particular, a fish population subject to various degrees and types of fishing. The differential equation models are given below. (Several possible choices are listed in Table 1.10.) In this report, a discussion of the meaning of each variable and parameter and an explanation of why the equation is written the way it is were done. The three general approaches that can be employed to study a differential equation: Numerical techniques yield graphs of approximate solutions, geometric/qualitative techniques provide predictions of the long-term behavior of the solution and in special cases analytic techniques to provide explicit formulas for the solution were employed if necessary. In this report, many of these techniques were done to help the understand the models. The following equations were considered:

1. (Logistic growth with constant harvesting) The equation

$$\frac{dp}{dt} = kp\,(1 - \frac{p}{N}) - a$$

represents a logistic model of population growth with constant harvesting at a rate a. For a = a_1, what will happen to the fish population for various initial conditions? (Note: This equation is autonomous, so you can take advantage of the special techniques that are available for autonomous equations.)

2. (Logistic growth with periodic harvesting) The equation

$$\frac{dp}{dt} = kp\cdot\left(1 - \frac{p}{N}\right) - a(1 + \sin(bt))$$

is a non-autonomous equation that considers periodic harvesting. What do the parameters a and b represent? Let b = 1. If a = a1, what will happen to the fish population for various initial conditions?

3. Consider the same equation as in Part 2 above, but let a = a_2. What will happen to the fish population for various initial conditions with this value of a?

In this report the three questions were addressed one at a time, in the form of a short essay. It began with Questions 1 and 2 with a description of the meaning of each of the variables

and parameters and an explanation of why the differential equation is the way it is. Pictures and graphs of data and of solutions of models were included as appropriate.

IV. RESULTS AND DISCUSSION

A. Logistic Growth with Constant Harvesting

Constant harvesting is where a fixed number of fish were removed from the stock at constant time rate (Gilbert, 2013). We assume that a constant number, *a*, of fish are removed from the population. Hence, the model is given by

$$p' = \frac{dp}{dt} = kp\left(1 - \frac{P}{N}\right) - a \tag{3}$$

where p = size of the population (in this example, fish population)

t = time

k = per capita growth rate

N = carrying capacity of the environment

a = constant number of fish harvested each time.

Setting the equation (3) equal to zero, we have two equilibrium equations. These occurs when the growth rate of the fish population is equivalent to the harvest rate, thus,

$$\frac{dp}{dt} = kp\left(1 - \frac{p}{N}\right) - a \tag{3}$$

$$\int \frac{dp}{kp\left(1 - \frac{p}{N}\right) - a} = \int dt \tag{4}$$

$$\int \frac{dp}{kp\left(1 - \frac{p}{N}\right) - a} = \int \left(\frac{A}{p - p_1} + \frac{B}{p - p_2}\right) dt \tag{5}$$

$$\int \frac{dp}{kp\left(1 - \frac{p}{N}\right) - a} = A \ln|p - p_1| + B \ln|p - p_2| \tag{6}$$

$$\int \frac{dp}{kp\left(1 - \frac{p}{N}\right) - a} = \ln\left|(p - p_1)^A (p - p_2)^B\right| \tag{7}$$

which implies that

$$(p - p_1)^A (p - p_2)^B = Ce^t \tag{8}$$

where p_1 and p_2 are roots to the quadratic polynomial in p and by partial fractions, $A=(p_1-p_2)^{-1}$, $B=(p_2-p_1)^{-1}$. Assuming $p(0)=p_0$, we get $C=(p_0-p_1)^A(p_0-p_2)^B$. Solving for p explicitly requires the values for N, k, and a to find p_1, p_2, A and B.

In solving the differential equation, equilibrium solutions should be first solved.

If $\frac{dp}{dt}=0$;

$$kp\left(1-\frac{p}{N}\right)-a=0 \tag{9}$$

$$\frac{-N}{p}\left(kp+\frac{kp^2}{N}+a\right)=0 \tag{10}$$

$$p^2-Np+\frac{aN}{k}=0 \tag{11}$$

where:

a=1

b=-N

c=$\frac{aN}{k}$

To get the equilibrium solutions,

$$p(t)=\frac{-b\pm\sqrt{b^2-4ac}}{2a} \tag{12}$$

$$p_1(t)=\frac{-(-N)\pm\sqrt{N^2-4(1)\left(\frac{aN}{k}\right)}}{2(1)} \tag{13}$$

$$p_1(t)=\frac{N+\sqrt{N^2-4\left(\frac{aN}{k}\right)}}{2} \tag{14}$$

$$p_2(t)=\frac{N-\sqrt{N^2-4\left(\frac{aN}{k}\right)}}{2} \tag{15}$$

The value of $kN \neq 4a$ otherwise $p_1=p_2$. Note that p_2 is not possible for $\sqrt{N^2 - 4\frac{aN}{k}} > N$ so cases $p_1(t) > p_2(t)$ for all t will be considered. We define $p_1=p^+$ and $p_2=p^-$. Linearization is applied.

$$\left.\frac{df}{dp}\right|_{p-p\pm} = k\left(1 - \frac{2p \pm}{N}\right) \tag{16}$$

$$\left.\frac{df}{dp}\right|_{p-p\pm} = k\left(\pm\sqrt{1 - \frac{4a}{kN}}\right) \tag{17}$$

which implies that $p_1=p^+$ is a sink and $p_2=p^-$ is a source (Zill).

Table 1 The chosen initial condition and the solved p1 and p2

Choice	k	N	a1	p1	p2
3	0.2	5	0	5	0
			0.21	3.5	1.5
			0.22	3.36603	1.63397
			0.23	3.20711	1.79289
			0.24	3	2
			0.25	2.5	2.5

The above table has value for $a1$ is set from 0 to 0.25, where 0.25 represent the maximum limit. Extending the value will result to the equation obtaining an undefined answer. In the table, equations (16) and (17) were used to solve for the value of p_1 and p_2 respectively.

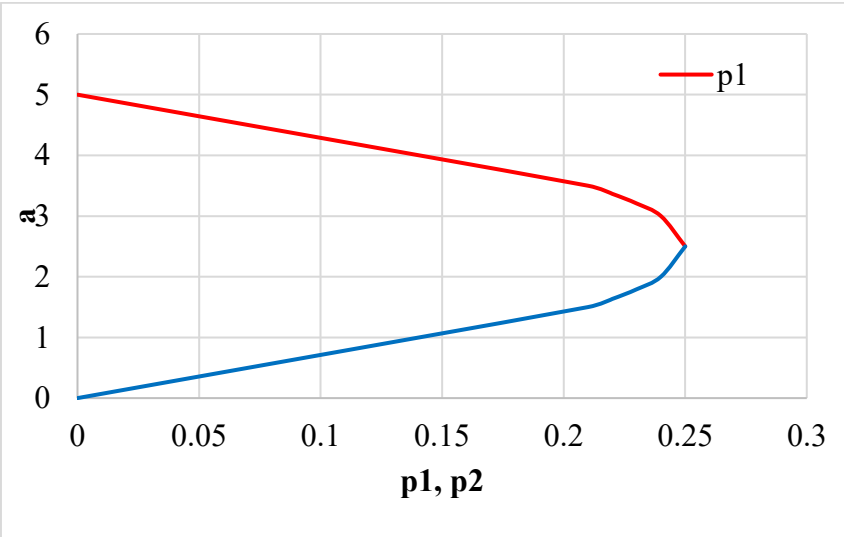

Figure 1 Graph of Table 1

The above figure shows an increase in p_2 and a decrease in p_1 which implies stability.

B. Logistic Growth with Periodic Harvesting

Functions or phenomena that repeat or are cyclic in nature are model with sine and cosine functions with a period of 2π. When fish population is harvested periodically, then the harvesting rate model becomes a nonautonomous equation as

$$A(t) = a(1 + \sin(bt)) \tag{18}$$

where a represents the coefficient that determines the total rate of periodic harvesting and additional 1 is included to insure a positive quantity for sine term.

Hence, we modify the logistic growth model (2) to include periodic harvesting, we have

$$\frac{dp}{dt} = kp \cdot \left(1 - \frac{p}{N}\right) - a(1 + \sin(bt)) \tag{19}$$

where

p = size of the fish population

k = growth rate of fish

N = carrying capacity of the environment

a = coefficient that determines the total rate of periodic harvesting

b = period of the sine function

Equation (11) might model extinction for stocks less than some threshold population P_0, and a stable population that oscillates about carrying capacity N with period equal to $T = \frac{2\pi}{b}$ (Gilbert, 2013).

In getting the solutions of periodic harvesting usual methods from calculus can no longer be applied since it is a non-separable differential equation. It is a nonautonomous nonlinear differential equation periodically forced with period $T = 1$ and depends explicitly on time. Thus, an online graphing tool for differential equations by Bluffton University were employed – a more qualitative approach to nonautonomous nonlinear differential equations (Nester, 2018).

For initial condition $a=a_1$, values of k and N be of choice 3, then equation *(19)* will be

$$\frac{dp}{dt} = 0.20\,p\left(1 - \frac{p}{5}\right) - 0.21(1 + \sin(t)) \tag{20}$$

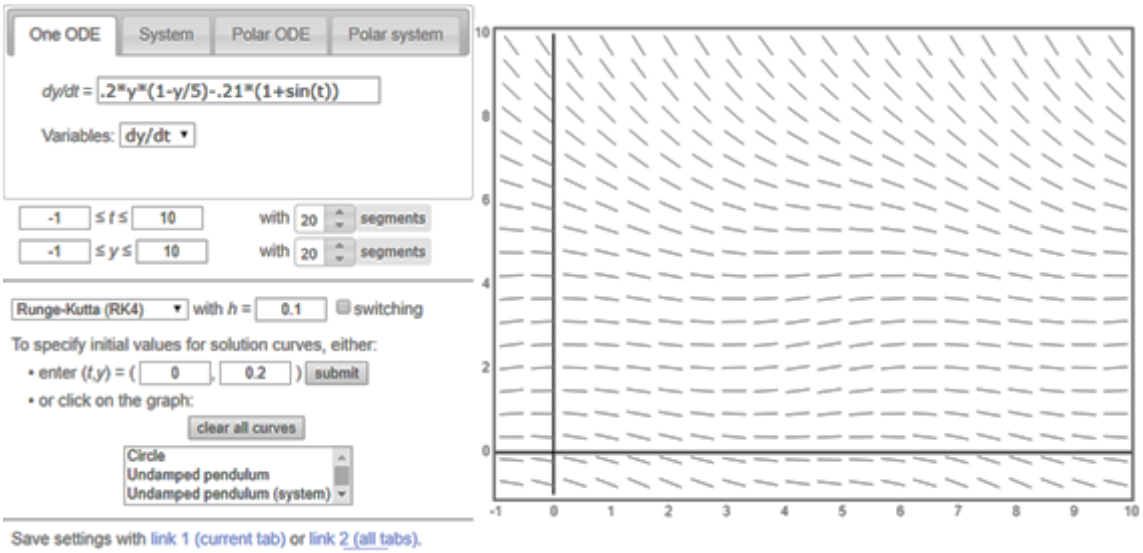

Figure 2 Slope Field of Equation (20)

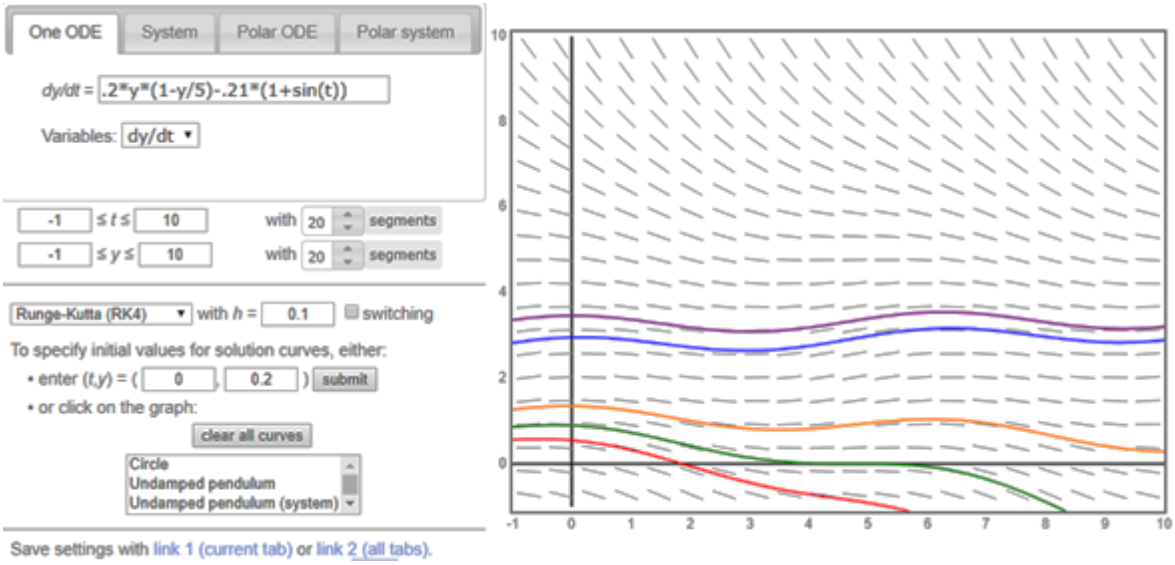

Figure 3 Phase Line for Equation (20)

The above figure displays the phase line solution of the given differential equation and its different values. The graphs are of wave functions than indicate the rapidly decreasing fish population.

At $a=a_2$, the differential equation becomes

$$\frac{dp}{dt} = 0.20\,p\left(1 - \frac{p}{5}\right) - 0.25(1 + \sin(t)) \tag{21}$$

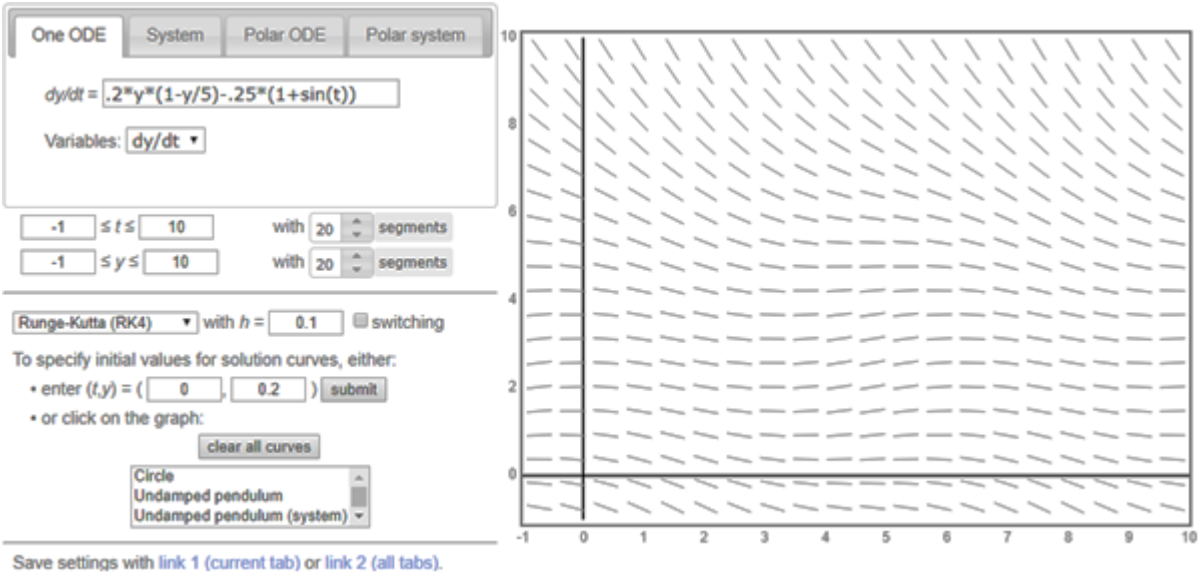

Figure 4 The slope field of equation of Equation (21)

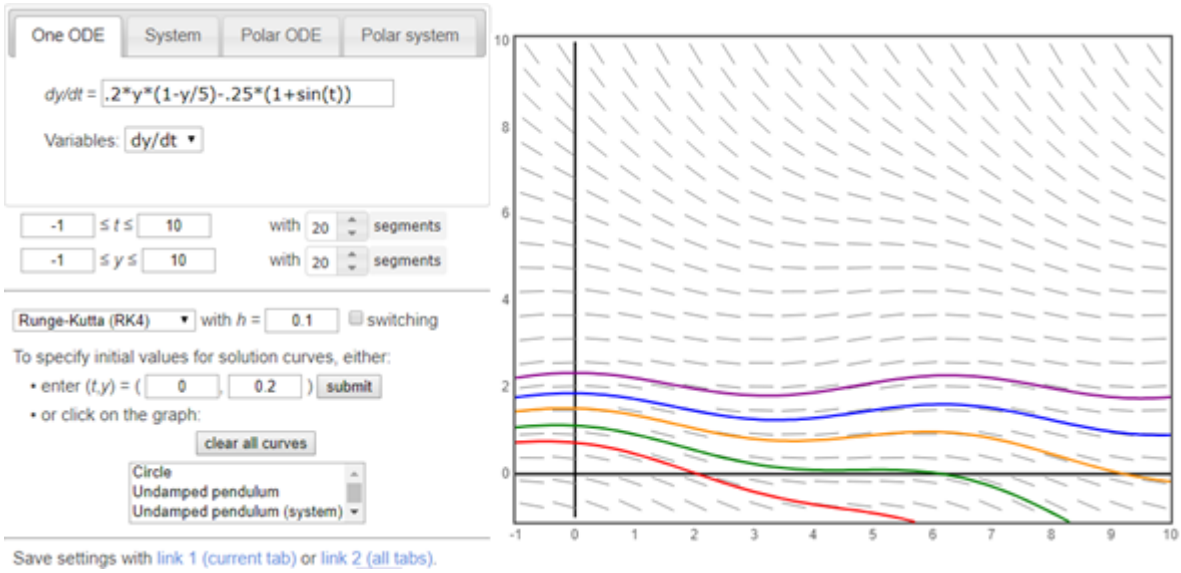

Figure 5 Phase Line for Equation (21)

The figure shows the phase lines for the solution of Equation (21) at different values. The sine function is seen apparent and it shifts the equilibrium state below the carrying capacity.

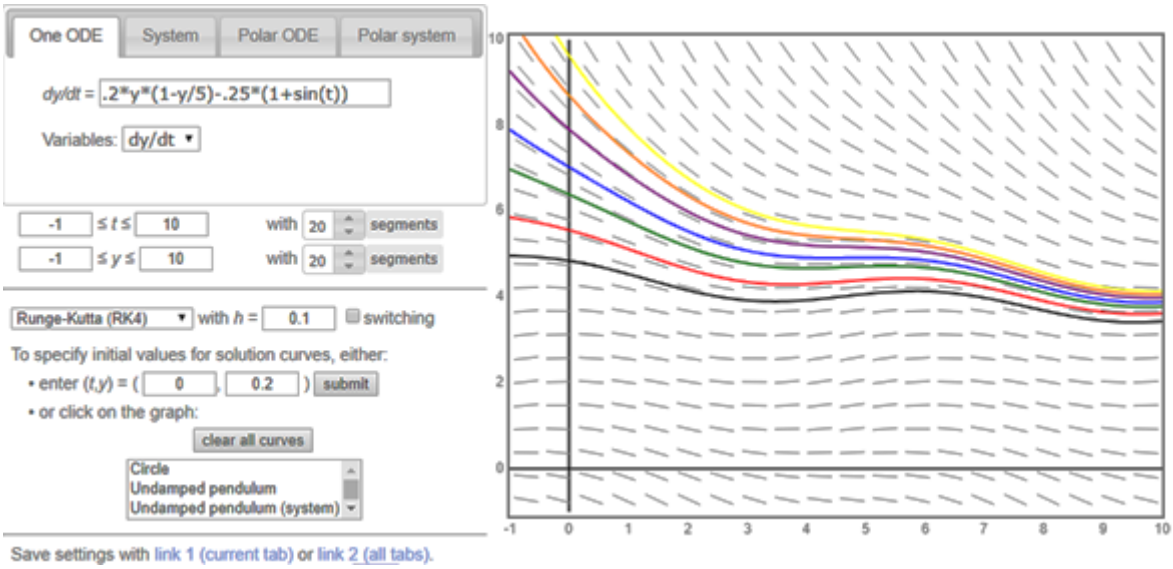

Figure 6 Phase Lines after the values of 0.25

For all initial values, the equilibrium states disappear and the population rapidly drops off.

V. CONCLUSION

The laboratory experiment showed the modified logistics equation to appropriate for a fish population with constant harvesting and with periodic harvesting. These models showed the direct relationship of harvesting rate and long-term population: increasing harvesting would lead to more extinction trajectories. The first model is an autonomous system that has equilibrium solutions. The second one is a non-autonomous system which has 'steady-state solutions' that could be used as reference point to describe the behavior of neighboring solutions. Considering these long term patterns as steady-state solutions, it can then be concluded that the effect of harvesting is to shift these steady-state solutions. Equilibrium solutions as well as steady-state solutions make it possible to determine the long-term behavior of a particular solution satisfying an initial condition.

VI. APPENDIX

Table 2 Possible Choices for the parameters

Choice	k	N	a_1	a_2
1	0.25	4	0.16	0.25
2	0.50	3	0.21	0.25
3	0.20	5	0.21	0.25
4	0.20	5	0.16	0.25
5	0.25	4	0.09	0.25
6	0.20	5	0.09	0.25
7	0.50	2	0.16	0.25
8	0.20	5	0.24	0.25
9	0.25	4	0.21	0.25
10	0.50	2	0.09	0.25

VI. References

Blanchard, P., Devaney, R. L., & Hall, G. R. (2010). *Differential Equations* (4th ed.). U.S.A.: Brooks/Cole.

Gilbert, A. (2013). *Modelling Sustainable Harvesting Strategies of a Fish Pond - A Case Study.* Kumasi: Department of Mathematics, Kwame Nkrumah University of Science and Technology.

Nester, D. (2018). *Slope and Direction Fields*. Retrieved from Boston University: https://www.bluffton.edu/homepages/facstaff/nesterd/java/slopefields.html

Zill, D. G. (n.d.). *A First Course in Differential Equations with Modeling Applications* (9th ed.).